

Problem Set

1. Normal–Normal Model (Posterior, Estimation, Interval)

Let

$$X_1, \dots, X_n \sim N(\mu, \sigma^2), \quad \sigma^2 \text{ known}$$

and prior

$$\mu \sim N(\theta, \tau^2).$$

- (a) Derive the posterior distribution of $\mu \mid x$.
- (b) Compute the posterior mean and variance.
- (c) Show that the posterior mean is a weighted average of $\bar{x}$ and θ .
- (d) Obtain a $(1 - \alpha)$ credible interval for μ .
- (e) Compare this with the frequentist confidence interval and explain the difference in interpretation.

2. Improper Prior and Its Effect

Consider the same model as above, but take the improper prior:

$$\pi(\mu) \propto 1.$$

- (a) Derive the posterior distribution.
- (b) Show that the posterior mean equals the sample mean.
- (c) Explain why the posterior depends only on the likelihood.
- (d) Compare with the result from the previous problem.

3. Binomial–Beta Model (Posterior and Sequential Updating)

Let

$$X \sim \text{Binomial}(n, p), \quad p \sim \text{Beta}(\alpha, \beta).$$

- (a) Derive the posterior distribution of $p \mid x$.
- (b) Compute the posterior mean.
- (c) Show that

$$E[p|x] = w \cdot \frac{x}{n} + (1 - w) \cdot \frac{\alpha}{\alpha + \beta}$$

and find w .

- (d) Suppose data arrive in two stages:
 - x_1 successes in n_1 trials
 - x_2 successes in n_2 trials

Show that sequential updating gives the same posterior as using all data at once.

4. Bayes Estimators under Different Loss Functions

Let

$$\pi(\theta|x) = \text{Beta}(a, b).$$

- (a) Find the Bayes estimator under squared loss.
- (b) Find the MAP estimator.
- (c) Compare these estimators and comment on when they coincide.

5. Posterior Variance and Prior Influence

Using the Binomial–Beta model:

- (a) Write down the posterior variance.
- (b) Show that as $n \rightarrow \infty$, the posterior variance tends to 0.
- (c) Explain how the influence of the prior changes with increasing sample size.

6. Credible and HPD Intervals

- (a) Define a $(1 - \alpha)$ credible interval.
- (b) Define the HPD interval.
- (c) Show that for a normal posterior distribution, the HPD interval coincides with the equal-tail credible interval.
- (d) Explain why this is not generally true for a Beta posterior.

7. Markov Chain (3-State Case)

Consider the transition matrix

$$P = \begin{bmatrix} 1-2a & a & a \\ a & 1-2a & a \\ a & a & 1-2a \end{bmatrix}, \quad \text{where } 0 < a < \frac{1}{2}.$$

- (a) Compute P^2 .
- (b) Show by induction that

$$P^n = (1-3a)^n I + (1 - (1-3a)^n) \frac{1}{3} \mathbf{1}\mathbf{1}^T,$$

where $\mathbf{1} = (1, 1, 1)^T$.

- (c) Find the stationary distribution π .
- (d) Using the expression for P^n , show that

$$P^n \rightarrow \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix} \quad \text{as } n \rightarrow \infty.$$

- (e) For what values of a does the convergence fail? Explain.

8. Waiting Time for Consecutive Sixes

A fair six-sided die is rolled repeatedly. Let T_k be the number of rolls required to obtain k consecutive sixes for the first time, and let $E_k = \mathbb{E}[T_k]$.

- (a) Consider the case $k = 2$. Model the problem using a Markov chain with states corresponding to the number of consecutive sixes observed so far, and compute E_2 .
- (b) By defining suitable states, and arguing similarity be show that for $k \geq 2$:

$$H_0 - H_i = \frac{6^{i+1} - 6}{5} \quad \forall i \leq k$$

where H_i denotes the expected time to get k six consecutive sixes when you already have i consecutive sixes.

- (c) Use the above to find out an expression of E_k .
- (d) Verify your formula for $k = 1$ and $k = 2$.
- (e) Can you generalize the result for a biased die where $P(6) = p$.

9. Optimal Proposal within a Parametric Family

Consider the target density

$$f(x) = e^{-x^2/2}, \quad x \in \mathbb{R}$$

and a proposal density

$$g(x) = \text{Laplace}(0, b).$$

- (a) Derive an expression for

$$M(b) = \sup_x \frac{f(x)}{g(x)}.$$

- (b) Analyze how $M(b)$ depends on b , and determine the value of b that minimizes $M(b)$.
- (c) Interpret your result in terms of efficiency of rejection sampling.

10. Proposal Mismatch and Mixture Design

Let

$$f(x) = 0.5\mathcal{N}(-3, 1) + 0.5\mathcal{N}(3, 1).$$

- (a) Suppose $g(x) = \mathcal{N}(0, \sigma^2)$. Derive

$$M(\sigma) = \sup_x \frac{f(x)}{g(x)}.$$

- (b) Identify where the ratio is maximized and explain why.
- (c) Discuss why a single Gaussian proposal is inefficient.
- (d) Propose an improved mixture proposal and explain how it reduces M .

11. Adaptive Envelope Estimation

Suppose

$$\hat{M} = \max_{i \leq n} \frac{f(X_i)}{g(X_i)}, \quad X_i \sim g.$$

- (a) Explain why $\hat{M}$ may underestimate the true M .
- (b) Discuss the consequences on correctness.
- (c) Suggest strategies to mitigate this issue.

12. Failure and Fix via Transformation

Let

$$f(x) = xe^{-x}, \quad x > 0.$$

- (a) Let $g(x) = e^{-x}$. Compute $\frac{f(x)}{g(x)}$ and show no finite M exists.
- (b) Let $y = \log x$. Derive $f_Y(y)$.
- (c) Consider

$$g_Y(y) = \frac{\lambda}{2} e^{-\lambda|y|}.$$

Compute $\frac{f_Y(y)}{g_Y(y)}$.

- (d) Show a finite M exists when $\lambda \geq 2$, and explain why.

13. Finite Markov Chain Analysis

State space $\{1, 2, 3, 4\}$ with transitions:

- $1 \rightarrow 2$ with probability 1,
- $2 \rightarrow 3$ with probability 1,
- $3 \rightarrow 1$ with probability $\frac{1}{2}$ and $3 \rightarrow 4$ with probability $\frac{1}{2}$,
- $4 \rightarrow 3$ with probability 1.

- (a) Show irreducibility.
- (b) Show aperiodicity.
- (c) Find π such that $\pi P = \pi$.
- (d) Verify detailed balance does not hold.

14. MH vs Rejection Sampling

Let $q(y|x) = g(y)$.

- (a) Write the acceptance probability.
- (b) Show when it behaves like rejection sampling.
- (c) Explain the intuition.

15. Proposal Scale Effects

$$Y = x + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2).$$

- (a) Analyze behavior as $\sigma \rightarrow 0$.
- (b) Analyze behavior as $\sigma \rightarrow \infty$.
- (c) Discuss the trade-off.

16. Slice Sampling for Exponential-type Density

$$f(x) = xe^{-x}, \quad x > 0.$$

- (a) Write the joint density $\pi(x, u)$.
- (b) Derive $u \mid x$.
- (c) Describe sampling $x \mid u$.
- (d) Compare with rejection sampling.

17. Gibbs Sampling for a Bivariate Distribution

$$\pi(x, y) \propto \exp\left(-\frac{x^2}{2} - \frac{(y-x)^2}{2}\right).$$

- (a) Derive $\pi(x \mid y)$ and $\pi(y \mid x)$.
- (b) Identify their distributions.
- (c) Write the Gibbs sampler.
- (d) Explain why Gibbs is suitable here.